



Towards Error Mitigation Applications for Oxford Nanopore Technologies

Advisors: Zhiyi Sun

Laurence Ettwiller

Genome Biology Division

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3GS Tech Overview

Platform	PacBio RS	ONT MinION
Cost (\$ K)	695	MAP (1 + .270/run)
Size (in ³ / lbs)	176,000 / 1,895	12 / <2
Throughput (Gb)	0.5	0.05
Run time (hrs)	3-4	48-72
Read length (bp)	10K	8K
Observed error	~11% (single-pass)	>20%
Quality score	Q40	<Q10

ONT Mirrored

ONT-specific Applications

(++) portability

(+) cost

(+) direct interrogation on

(--) high error rates/

(--) size

- Clinical microbiology

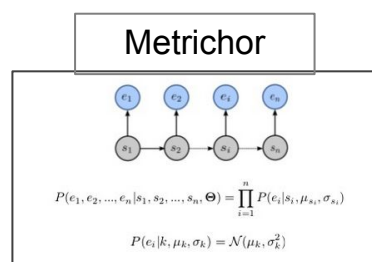
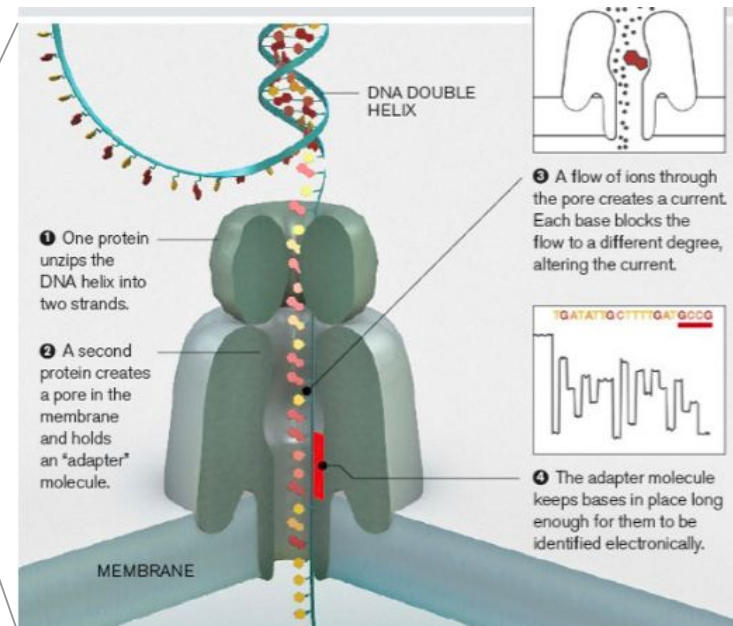
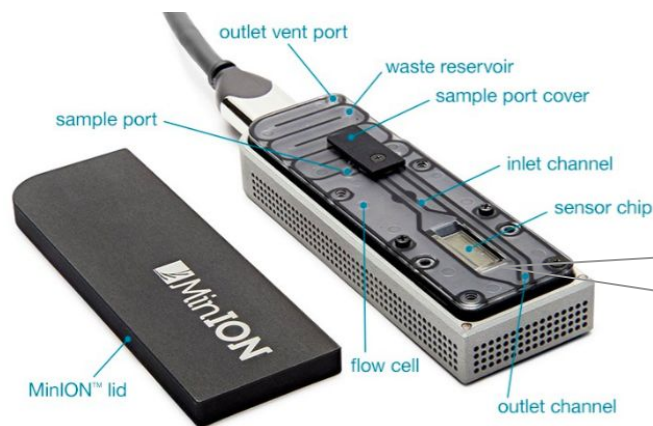
- Precision medicine

- In field *de novo* assembly

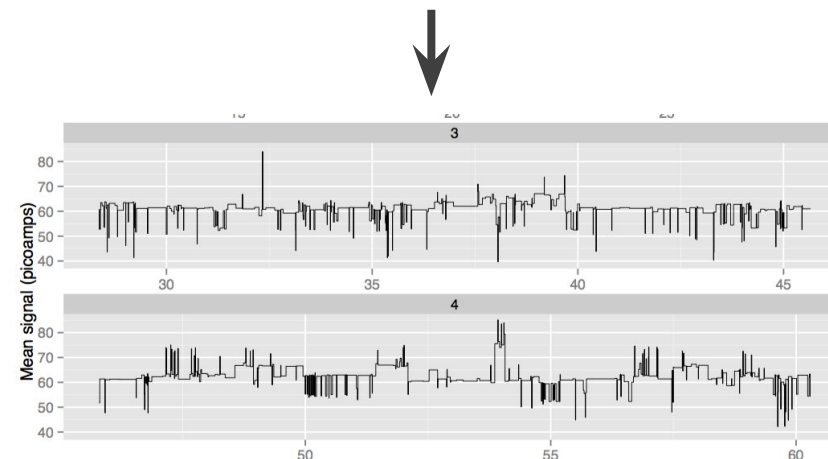
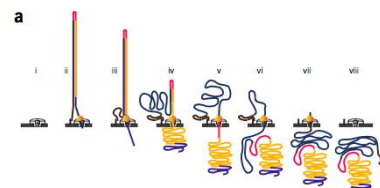
d - Epigenomics

- Structural variation analysis

ONT Sequencing Mechanism



...CGATC



Project Concept

Primary:

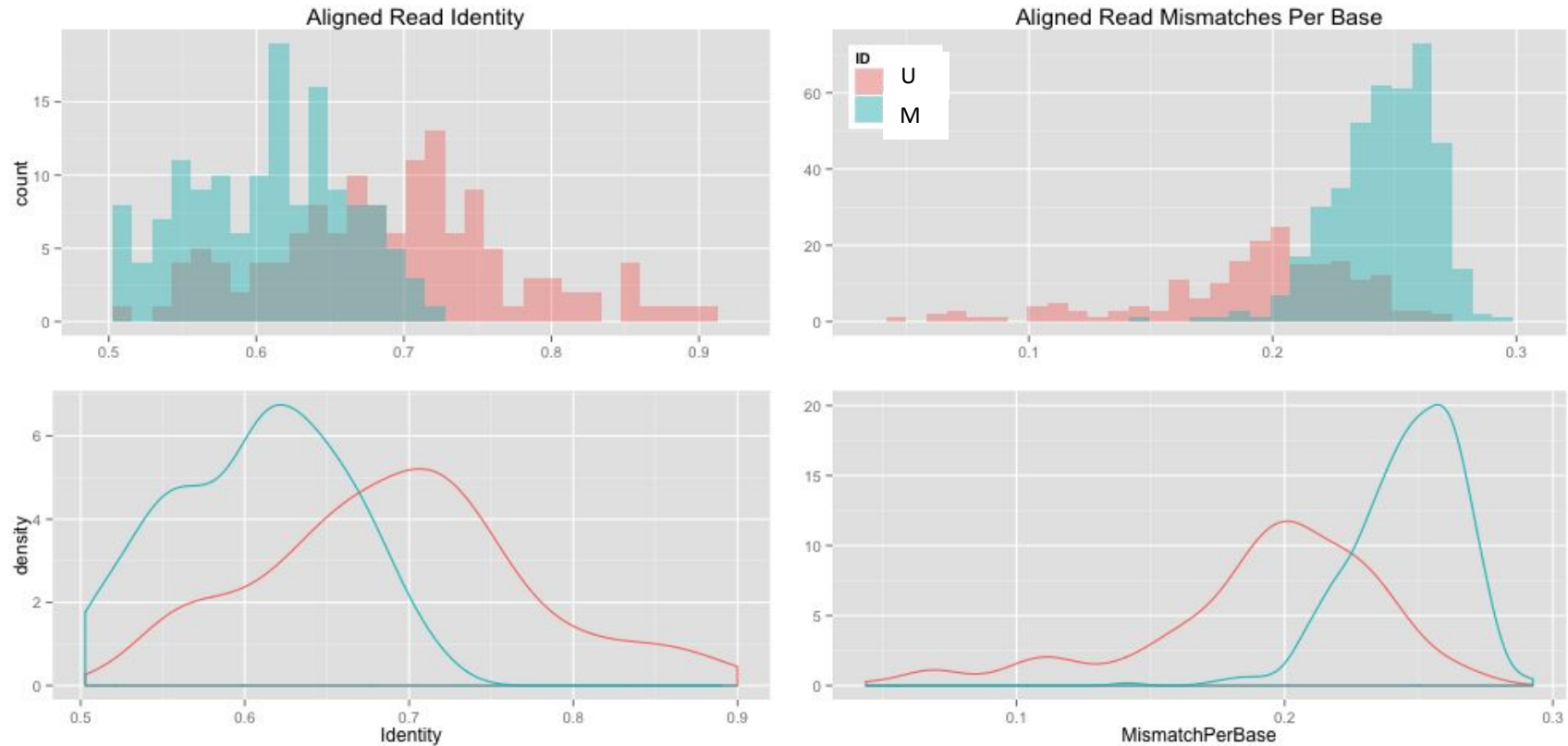
Use ONT to analyze modified DNA to determine feasibility of pre-sequencing modification of substrates for 'error mitigation'.

- (1) Do base modifications affect ONT read distributions?
- (2) Can modifications produce more easily distinguishable signal patterns?

Secondary:

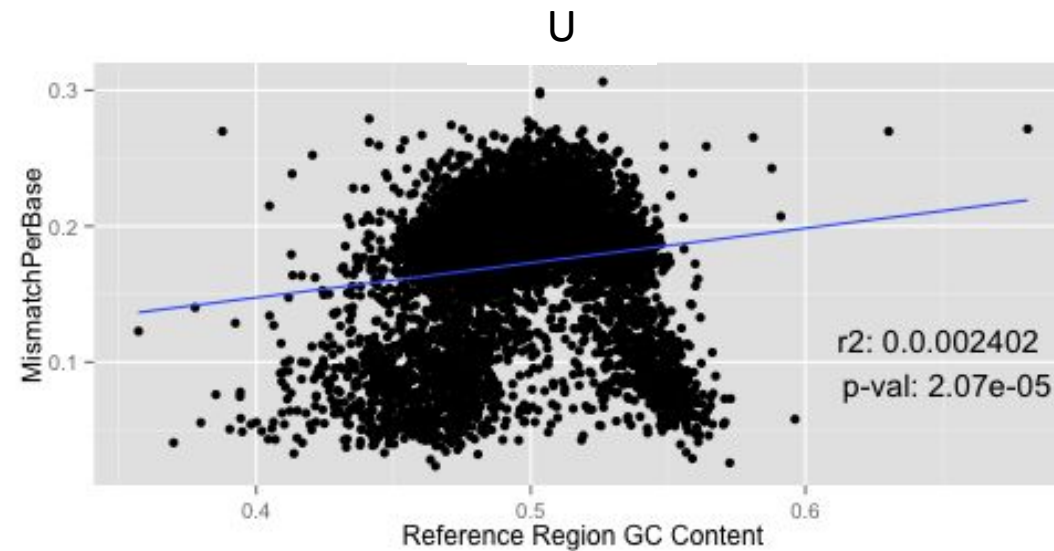
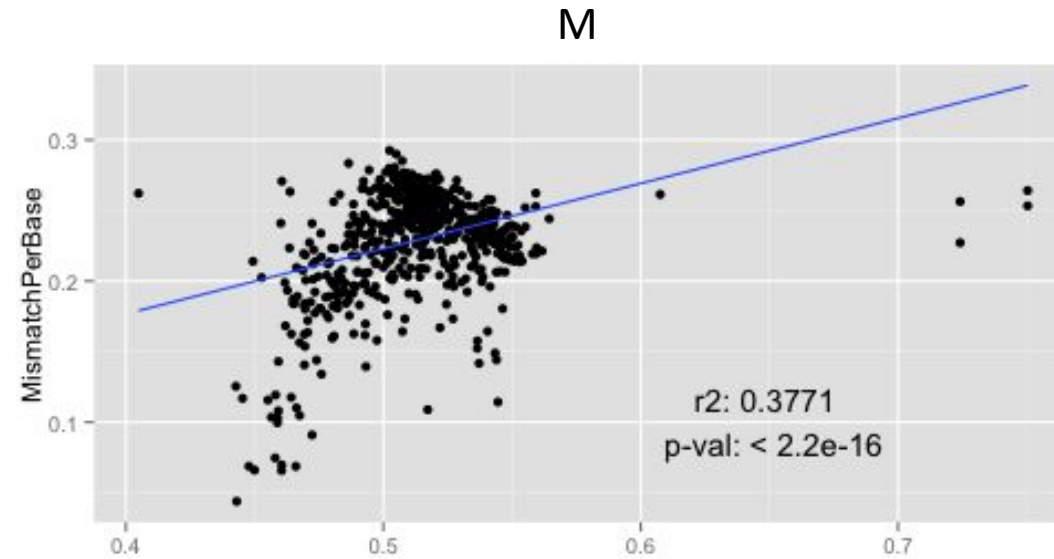
- Assess computational tools available for ONT data
- Build pipeline for future ONT data processing/ analysis

M and U give different distributions



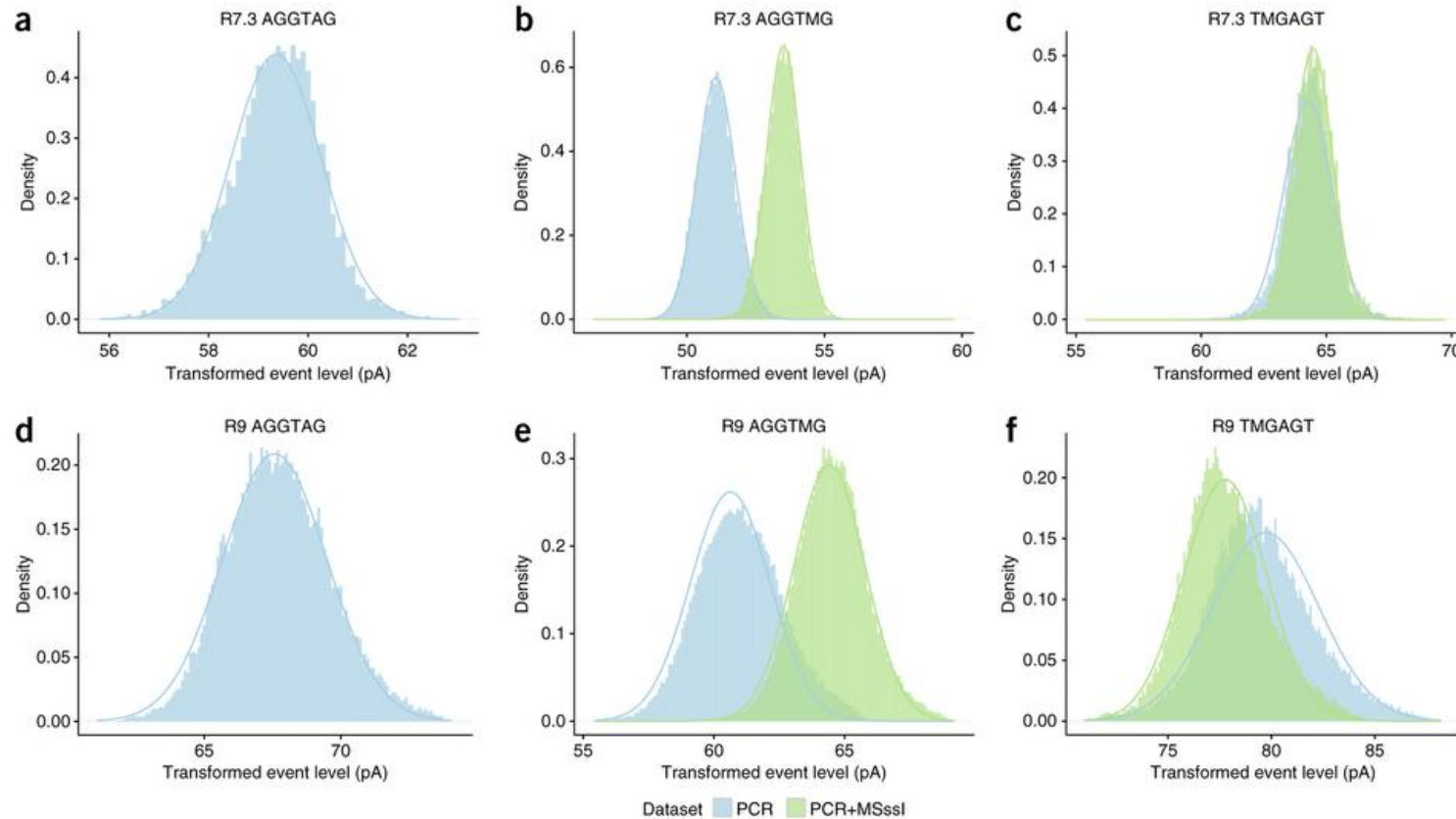
U	M	
55.6%	36.3%	Mean Identity
0.192	0.245	Mean MismatchPerBase

M positively correlated with error



More Recent Developments

Simpson JT, Workman RE, Zuzarte PC, David M, Dursi LJ, Timp W. 2017. **Detecting DNA cytosine methylation using nanopore sequencing.** *Nat Methods*, 14: 407–410. doi:10.1038/nmeth.4184.



More Recent Developments

Rand AC, Jain M, Eizenga JM,
Musselman-Brown A, Olsen HE,
Akeson M, Paten B. 2017.

**Mapping DNA methylation with
high-throughput nanopore
sequencing.** *Nat Methods*, 14:
407–410.

doi:10.1038/nmeth.4189.

